Temperature Coefficient of Resistance for Copper

The Temperature Coefficient of Resistance for Copper (near room temperature) is **0.393% per degree Celsius (C)** meaning that for every 1°C rise in temperature, the resistance increases by 0.393%.

The relationship can be expressed in the formula:

$R = R_{ref} [1 + \alpha(T - T_{ref})]$

Where:

R	=	Conductor resistance at temperature "T" in Ohms
Rref	=	Conductor resistance at reference temperature (Tref) in Ohms
α	=	Temperature coefficient of resistance for the conductor material
T	=	Conductor temperature in degrees C
Tref	=	Reference temp at which α is specified for the conductor material in degrees C

Example:

Assume 100 feet of 20 gauge wire has a resistance of 1.015 Ohms at 20°C (room temp). If the temperature of the wire increases by 10°C, what will the resistance of the wire be?

$R = Rref [1 + \alpha(T – Tref)]$

$R = 1.015 [1 + 0.00393 (30 – 20)]$

$R = 1.015 [1 + 0.00393 (10)]$

$R = 1.015 [1 + 0.0393]$

R = 1.0549 Ohms

The change in resistance (ΔR) can be expressed by a simple change to the original formula:

$\Delta R = Rref [\alpha(T – Tref)]$

Example:

Using the same assumptions as in the previous example, what will the change in the resistance of the wire be?

$\Delta R = Rref [\alpha(T – Tref)]$

$\Delta R = 1.015 [0.00393 (30 – 20)]$

$\Delta R = 1.015 [0.00393 (10)]$

$\Delta R = 1.015 [0.0393]$

ΔR = 0.0399 Ohms

The temperature coefficient of resistance for some common materials at 20°C:

- Copper = 0.00393
- Aluminum = 0.004308
- Iron = 0.005671
- Nickel = 0.005866
- Gold = 0.003715
- Tungsten = 0.004403
- Silver = 0.003819

Further details about temperature coefficients of resistance can be found on the All About Circuits (https://www.allaboutcircuits.com/textbook/direct-current/chpt-12/temperature-coefficient-resistance/) web site.

A calculator is available on the University of Georgia HyperPhysics (http://hyperphysics.phy-astr.gsu.edu/hbase/electric/restmp.html) web site.

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